

MNC Patrol Effort

User Guide

Configuring and running the workflow that turns EarthRanger patrol data into trajectory maps, effort summaries, and a coverage dashboard for Mara North Conservancy

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Workflow id: **mnc_patrol_effort**

Overview

This guide walks you through configuring and running the MNC Patrol Effort workflow, which processes patrol events and observations from EarthRanger to produce trajectory maps, patrol effort summaries, a patrol coverage analysis, and a results dashboard for Mara North Conservancy.

The workflow delivers, for each run:

- **Events summary** — total events recorded by date and by type, with a line chart (HTML + PNG)
- **Patrol purpose summary** — patrol count grouped by patrol purpose (CSV + table)
- **Patrol relocations** — full observation dataset as a GeoParquet file
- **Foot patrol report** — effort summary table (CSV) and coverage map (HTML + PNG)
- **Vehicle patrol report** — effort summary table (CSV) and coverage map (HTML + PNG)
- **Motorbike patrol report** — effort summary table (CSV) and coverage map (HTML + PNG)
- **Combined trajectories** — merged trajectory dataset (GeoParquet)
- **Overall patrol efforts** — per-ranger summary of patrols, distance, and duration (CSV + table)
- **Patrol coverage map** — 1 000 m grid-cell visit density map (HTML + PNG) with conservancy occupancy percentage (CSV + table)
- **Results dashboard** — the four coverage maps, the events chart, and the three summary tables assembled into a single dashboard view

Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with `patrol_info` events and associated patrol observations recorded for the analysis period
- Network access to **Dropbox**, so the workflow can download the MNC conservancy boundary and parcels files at runtime

Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field, then click **Add Template**.

```
https://github.com/wildlife-dynamics/mnc-patrol-effort.git
```

Step 2 — Configure the EarthRanger Connection

Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

Field	Description
Data Source Name	A label to identify this connection (e.g. Mara North Conservancy)
EarthRanger URL	Your instance URL (e.g. your-site.pamdas.org)

Field	Description
EarthRanger Username	Your EarthRanger username
EarthRanger Password	Your EarthRanger password

Note: Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **mnc-patrol-effort**. Click the card to open the workflow configuration form.

Step 4 — Configure Workflow Details, Time Range, and EarthRanger Connection

The configuration form has three sections on a single page.

Set workflow details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. reporting month or site)

Time range

Field	Description
Timezone	Select the local timezone (e.g. Africa/Nairobi UTC+03:00)
Since	Start date and time — all events and patrol data from this point are fetched
Until	End date and time of the analysis window

Connect to ER

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown (e.g. Mara North Conservancy).

Once all three sections are filled, click **Submit**.

Running the Workflow

Once submitted, the runner will:

- 1 Download the MNC community conservancy boundary and parcels GeoPackage files from Dropbox; repair invalid geometries on the conservancy boundary; filter it down to the Conservancy grazing zone (used later as the coverage AOI) and to Mara North Conservancy (used for the map extent); build styled map layers for the conservancy boundary and parcels.
- 2 Fetch all events from EarthRanger; extract the date from each timestamp; add a temporal index; exclude `distancecountwildlife_rep`, `distancecountpatrol_rep`, and `airstrip_operations` events; summarise the remainder by date and by type; draw a daily events line chart; save as `total_events_recorded_by_date.csv`, `total_events_recorded_by_type.csv`, and `total_events_recorded.html/.png`.
- 3 Filter `patrol_info` events; flatten event details; save as `patrol_events.csv`; rename fields (`patrol_id`, `participants`, `patrol_purpose`, `transport_type`); summarise patrol count by patrol purpose; save as `patrol_purpose_summary.csv`.
- 4 Drop records with no patrol ID; fill missing transport type with `Undefined`; explode the `patrol_id` column; fetch patrol records and patrol observations from EarthRanger; merge them with the patrol info; explode the `participants` column; process the result into relocations (filtering out sentinel coordinates); save as `patrol_relocations.geoparquet`.
- 5 Split relocations into three transport-type branches — `Foot`, `Vehicle`, and `Motorbike` — and convert each to trajectories using type-appropriate segment filters (foot patrols use a tighter speed/distance envelope than vehicle and motorbike).
- 6 For each patrol type: rename trajectory columns; summarise effort metrics (patrol count, distance km, duration hrs, average speed) by `patrol_type_value`; save effort CSV.
- 7 For each patrol type, and again for the combined dataset: build a 1 000 m patrol coverage grid over the conservancy boundary; classify visit counts into 5 equal-interval bins; apply the `RdYlGn` colormap; draw the coverage map; save as HTML, convert to PNG, and wrap it in a map widget.
- 8 Concatenate the foot, vehicle, and motorbike trajectories; summarise per-ranger effort (patrol count, distance, duration) from the combined dataset; fill missing participant names with `Undefined`; save as `overall_patrol_efforts.csv`.
- 9 Reproject the conservancy boundary and compute what percentage of it is covered by the overall patrol coverage grid; save as `patrol_coverage.csv`.
- 10 Render the events chart, the patrol purpose summary, the overall patrol efforts, and the conservancy occupancy as table/chart widgets, and assemble all four maps, the chart, and the three tables into a single results dashboard.
- 11 Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`.

Events Summary

File	Description
total_events_recorded_by_date.csv	Daily event counts (all types combined)
total_events_recorded_by_type.csv	Daily event counts broken down by event type
total_events_recorded.html / .png	Line chart of daily event counts

Patrol Purpose

File	Description
patrol_events.csv	Flattened patrol_info event details
patrol_purpose_summary.csv	Patrol count by patrol purpose
patrol_purpose_summary_table.html	Rendered HTML table backing the dashboard widget

Relocations

File	Description
patrol_relocations.geoparquet	Full patrol observation dataset with patrol metadata

Foot Patrols

File	Description
foot_patrol_efforts.csv	Patrol count, distance, duration, and average speed by patrol type
foot_patrol_map.html / .png	Foot patrol coverage grid map

Vehicle Patrols

File	Description
vehicle_patrol_efforts.csv	Patrol count, distance, duration, and average speed by patrol type
vehicle_patrol_map.html / .png	Vehicle patrol coverage grid map

Motorbike Patrols

File	Description
motorbike_patrol_efforts.csv	Patrol count, distance, duration, and average speed by patrol type
motor_patrol_map.html / .png	Motorbike patrol coverage grid map

Combined Trajectories and Overall Effort

File	Description
patrol_trajectories.geoparquet	Reprojected overall patrol coverage grid (foot + vehicle + motorbike combined)
overall_patrol_efforts.csv	Per-ranger summary of total patrols, distance km, and duration hrs
overall_patrol_efforts_table.html	Rendered HTML table backing the dashboard widget

Patrol Coverage

File	Description
overall_patrol_map.html / .png	1 000 m grid-cell visit density map, all patrol types combined
patrol_coverage.csv	Patrol occupancy percentage per conservancy region
patrol_coverage_table.html	Rendered HTML table backing the dashboard widget

Dashboard

The final dashboard step assembles eight widgets, in order: the foot, vehicle, motorbike, and overall patrol coverage maps; the total events chart; and the patrol purpose, overall patrol efforts, and conservancy occupancy tables.

Widget	Source
Foot Patrol Coverage Map	foot_patrol_map.html
Vehicle Patrol Coverage Map	vehicle_patrol_map.html
Motorbike Patrol Coverage Map	motor_patrol_map.html
Overall Patrol Coverage Map	overall_patrol_map.html
Total Events Recorded	total_events_recorded.html
Patrol Purpose Summary	patrol_purpose_summary_table.html
Overall Patrol Efforts	overall_patrol_efforts_table.html
Conservancy Patrol Occupancy	patrol_coverage_table.html

Note: Each table widget is sortable and filterable in place; downloading directly from the widget is disabled — use the corresponding CSV output file for that.